

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
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Section / Batch:	B.TECH AIML	Date:	18-07-20206

Instructions to Students

- Answer the following question. It carries 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1

Q2. Smart Traffic Signal Optimization System

A smart traffic management system processes vehicle flow data using a divide-and-conquer algorithm represented by

the recurrence $T(n) = 2T(n/2) + n$. Apply the Master Theorem to determine the time complexity of the algorithm.

Clearly identify the parameters a , b , and $f(n)$, and justify the appropriate case selection.

Analyze how the

performance of the system changes as the number of vehicles increases significantly in urban areas. Interpret the final

result using asymptotic notation and explain how it affects real-time traffic signal optimization.

Smart Traffic Signal Optimization System

Given Recurrence

Step 1: Identify the Parameters

The recurrence is in the form:

Where:

$$a = 2$$

$$b = 2$$

$$f(n) = n$$

Step 2: Calculate

Step 3: Compare and

Since,

this satisfies Master Theorem Case 2.

Step 4: Apply Master Theorem

For Case 2:

Therefore,

Performance Analysis

As the number of vehicles (n) increases significantly:

The running time increases at a rate of .

The algorithm scales efficiently compared to quadratic-time algorithms ().

Even with heavy traffic, processing time grows moderately, making it suitable for large urban traffic networks.

Asymptotic Result

Best Case (Ω):

Worst Case (O):

Tight Bound (Θ):

Effect on Real-Time Traffic Signal Optimization

The divide-and-conquer approach efficiently processes large amounts of vehicle flow data. The complexity ensures the system can update traffic signals quickly as traffic volume increases.

This enables faster congestion detection, improved signal timing, reduced vehicle waiting time, and better traffic flow in smart cities.

Final Answer

$$a = 2$$

$$b = 2$$

$$f(n) = n$$

Master Theorem Case: Case 2

Time Complexity:

Interpretation: The algorithm scales efficiently with increasing traffic, making it well-suited for real-time smart traffic signal optimization

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I AKSHAYA KUMAR CHANDRASEKAR(192425350)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Akshaya kumar

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____